	PC	Fundamentals of Signal Processing & Communication	3	1	0	4	4
	PC	Computer Organization & Microprocessor	3	0	0	3	3
	PC	Computer Organization & Microprocessor Laboratory	0	0	2	2	1
			17	**1**	**8**	**26**	**22**
Sem V	OE	Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3
	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Theory of Computation & Compiler Design	3	0	0	3	3
	PC	Theory of Computation & Compiler Design Laboratory	0	0	2	2	1
	PC	Digital Signal Processing	3	0	0	3	3
	PC	Digital Signal Processing Laboratory	0	0	2	2	1
	PC	RF Engineering	3	0	0	3	3
	PC	RF Engineering Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VI	OE	Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Operating Systems	3	0	0	3	3
	PC	Operating Systems Laboratory	0	0	2	2	1
	PC	Digital Communication	3	0	0	3	3
	PC	Digital Communication Laboratory	0	0	2	2	1
	PC	Computer Communication and Networking	3	0	0	3	3
	PC	Computer Communication and Networking Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 4 (3-0-0)	3	0	0	3	3
	PE	Program Elective 5 (3-0-0)	3	0	0	3	3
	PC	Software Engineering Methodology	3	0	0	3	3
	PC	Software Engineering Methodology Laboratory	0	0	2	2	1
	PC	Embedded Systems	3	0	0	3	3
	PC	Embedded Systems Laboratory	0	0	2	2	1
	PC	Digital CMOS and VLSI Design	3	0	0	3	3
	PC	Digital CMOS and VLSI Design Laboratory	0	0	2	2	1
		Seminar					1
			18	0	6	24	24
Sem VIII		**Course Name**	**Th**	**Tut**	**Pra**	**Hrs**	**Cr**
	Pro	Major/Comprehensive Project					**12**

Program Core Electives:

Semester	Software and Networks	Artificial Intelligence	Communication and Signal Processing	Embedded and VLSI
5 (program core elective1)	Web Technology	Optimization methods and Algorithms		Introduction to CMOS and Memory Technology
	Problem solving through Java			
6 (program core elective2, 3)	Cloud Architecture and Service	Machine Learning	Image Processing	Hardware Accelerated Computing
	Advanced Web Technology	Data warehousing & Mining	Statistical Signal Processing	
	Advanced Algorithm Design			
7 (program core elective4, 5)	Cryptography and Network Security	Computer Vision	Optical Communication	Autonomous systems
	Internet of Things	Deep Learning and Reinforcement learning	Modern Wireless Communications	Hardware Accelerated Computing
	Blockchain Technology			Real time Operating Systems
	Mobile Application Development			

6th Semester

<20IC303T >					Digital Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart basic knowledge of digital modulation and demodulation techniques.
2. To Understand coding and multiple access principles
3. Understand the basic principles of wireless communication systems.

UNIT I: DIGITAL BASEBAND COMMUNICATION TECHNIQUES	**10 Hrs.**
Review of Analog and digital Messages, Low pass sampling, Aliasing, Interpolation, PAM, PPM and PWM, PCM: Quantization, Uniform and non-uniform quantization, Quantization noise, Companding laws, DPCM, DM ADM, and SDM, Time division multiplexing (TDM), Line coding, Pulse shaping, Digital receivers, Regenerative repeaters.	
UNIT II: CARRIER COMMUNICATION TECHNIQUES	**10 Hrs.**
Carrier Communication Techniques: ASK, FSK, PSK, QAM, DPSK, MSK, GMSK, M-ary, Coherent and non-coherent detection, Carrier synchronization, Frequency Division Multiplexing (FDM), Noise performance and comparison of digital communication systems.	
UNIT III: INFORMATION THEORY AND SOURCE CODING	**12 Hrs.**
Uncertainty and Information, Average Mutual Information and Entropy, Measure of Information, Source Coding Theorem, Huffman and Shanon-Fano Coding, Uniqueness Property, Channel Models, Channel Capacity, Error free Communication over a Noisy Channel, Information Capacity, Shannon Limit.	
UNIT IV: ERROR CONTROL CODING (CHANNEL CODING)	**10 Hrs.**
Selection of Channel Codes, Error Correcting Codes: Linear Block Codes, Hamming Distance, Perfect Codes and Hamming Codes, Cyclic Codes, Burst Error Correcting and Detecting Code, Introduction to CRC, BCH, Reed-Solomon, Convolution Codes.	
	Total: 42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to:

CO1	:	Remember principles of various digital communication methods.
CO2	:	Understand digital modulation, detection, coding, digital transmission and access techniques.
CO3	:	Apply mathematical concepts to model digital communication system.
CO4	:	Analyze digital communication systems to obtain various parameters like bandwidth, data rate etc.
CO5	:	Evaluate and compare performance of digital communication systems.
CO6	:	Design a block diagram level simple digital communication system.

TEXT/REFERENCE BOOKS

1. B. P. Lathi, Zhi Ding “Modern Digital and Analog Communication Systems”, Oxford University Press.
2. P. Chakrabarti, “Analog and Digital Communication”, Dhanpat Rai & Co.
3. Wayne Tomasi “Electronic Communications Systems”, Pearson education India.
4. Ranjan Bose, “Information Theory, Coding and Cryptography”, PHI.
5. Simon Haykins & Michael Moher, Communication Systems, 5th Edition, John Willey, India Pvt. Ltd, 2010.
6. Simon Haykins & Michael Moher, An Introduction to Analog and Digital Communication, John Wiley India Pvt. Ltd., 2008.
7. T. Rappaport, “Wireless Communications – Principles and Practice”, Prentice Hall.

24IC***P					Digital Communication Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart knowledge of digital modulation, demodulation and coding techniques.
2. To verify performance of various baseband and carrier digital communication systems.
3. To compare various digital communication systems and understand their applications.

LIST OF EXPERIMENTS	
1	PAM, PPM, PWM baseband techniques
2	Sampling, Quantization, PCM and variants
3	Digital carrier communication (ASK, FSK, PSK etc.) techniques
4	TDM techniques and synchronization
5	QAM and FDM Techniques
6	Error detection and correction techniques
7	Noise and BER analysis in various digital communication systems
8	OFDM Techniques
9	Spread spectrum communication

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember principles of various digital communication methods.
CO2	:	Understand digital modulation, detection, coding, digital transmission and multiplexing techniques.
CO3	:	Apply theoretical and mathematical concepts to practical digital communication system.
CO4	:	Analyze and report the characteristics of various modulation, demodulation, multiplexing, synchronization, coding techniques.
CO5	:	Evaluate and report performance of digital communication systems.
CO6	:	Design and implement small scale digital communication system using development tools.

TEXT/REFERENCE BOOKS

1. B.P.Lathi, Zhi Ding "**Modern Digital and Analog Communication Systems**", Oxford University Press.
2. P. Chakrabarti, "**Analog and Digital Communication**", Dhanpat Rai & Co.
3. Wayne Tomasi "Electronic Communications Systems", Pearson education India.
4. Ranjan Bose, "**Information Theory, Coding and Cryptography**", PHI.
5. T. Rappaport, "**Wireless Communications – Principles and Practice**",Prentice Hall.

24ICT***T					Computer Communication and Networking					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn the fundamentals of networking layers and protocols.
2. To be able to understand, analyze, and design computer communication systems.
3. To develop various networking algorithms for enhancing the efficiency of existing networking protocols.
4. To study real-life applications of different networking protocols.

UNIT I: PHYSICAL AND DATA LINK LAYERS	**12 Hrs.**
Introduction: Nuts and Bolts, Performance parameters: throughput, delay, etc., Layered Architecture (OSI and TCP/IP), Data link layer: Introduction, Media access protocols (ALOHA, CSMA based) , Ethernet 802.3, Token, ring 802.5, Reliability Issue: sliding window, Internetworking and Routing: Best effort Service, Virtual Circuits, IP Addressing, introduction to physical layer services, different types of modulation schemes, multiplexing and de-multiplexing techniques, parity, cyclic redundancy check.	
UNIT II: NETWORK LAYER	**12 Hrs.**
Internetworking and Routing: Routing Issues, Distance Vector and Link State routing, Intra and Inter Autonomous System Routing (OSPF, RIP, BGP), Broadcast and Multicast Routing Issues. Network layer data plane: Concept of data forwarding, Longest prefix matching, switching fabrics, packet scheduling, IP addressing: IPV4 and IPV6, concept of subnet, NAT.	
UNIT III: TRANSPORT LAYER	**10 Hrs.**
Transport Layer: End to end delivery issues, Reliable data transfers, Congestion Control, Traffic engineering and Quality of service, TCP, UDP, Advanced Topics: QoS over IP, IPV6, and Infrastructure-less networks: wireless ad hoc and sensor networks, and Internet of Things (IoT), Green computer networking.	
UNIT IV: APPLICATION LAYER	**8 Hrs.**
Network application Design, Socket Programming, Client-server applications, Echo and Chat applications, DHCP, HTTP, FTP, DNS, Peer to Peer file sharing application.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember concepts of protocol design at different network layers
CO2	:	Understand design and implementation of different networking protocols
CO3	:	Apply computer networking concepts to solve problems
CO4	:	Analyze computer networking protocols at different layers
CO5	:	Evaluate performance of different computer network protocols
CO6	:	Create computer networking applications

TEXT/REFERENCE BOOKS

1. James Kurose and Keith Rose, **"Computer Networking: A Top Down Approach",** Pearson Education
2. Larry L Peterson and Bruce S Davie, **"Computer Networks: A Systems Approach",** Elsevier
3. Andrew S Tanenbaum, **"Computer Networks",** Pearson Education
4. Behrouz A Forouzan, **"Data Communication and Networking",** McGraw Hill

24ICT***P					Computer Communication and Networking Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To learn the fundamentals of networking layers and protocols.
2. To be able to analyze and design computer communication systems.
3. To be able to implement various networking algorithms and analyze the performance using network simulators.

LIST OF EXPERIMENTS	
1	To study and prepare LAN cables (cross and straight), to configure LAN and perform Static Routing
2	Introduction to Socket Programming- Design and Implement client-server elements of a few network applications e.g. TCP/UDP client server, Echo client and server, Time client and server, Online Quiz and Buzzer Application, etc
3	Configure DHCP in a small LAN and understand its functionality using Wireshark/ Packet Tracer
4	Configure DNS in a small LAN and understand its functionality using Wireshark/ Packet Tracer
5	Understand functionality of HTTP using Wireshark/ Packet Tracer
6	Understand functionality of TCP and UDP using Wireshark/ Packet Tracer
7	Configure virtual LAN and understand its functionality using Wireshark/ Packet Tracer
8	Configure a wired network with Hosts, switches, hubs and routers using Packet Tracer
9	Configure a wireless LAN (Wi-Fi) using Packet Tracer.
10	Simulation of TCP/UDP connections and performance analysis
11	Simulation of a vehicular network using network simulator (NS3/NETSIM) and performance analysis
12	Simulation of 5G network using network simulator (NS3/NETSIM) and performance analysis

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember concepts of functionality of layered network architecture.
CO2	:	Understand various types of network topologies, network devices and their functions within a network.
CO3	:	Apply computer networking concepts to solve problems.
CO4	:	Analyze computer networking scenarios using network simulator tools.
CO5	:	Evaluate different computer network protocols practically.
CO6	:	Create computer networking applications and scenarios.

TEXT/REFERENCE BOOKS

1. James Kurose and Keith Rose, **"Computer Networking: A Top Down Approach",** Pearson Education
2. Larry L Peterson and Bruce S Davie, **"Computer Networks: A Systems Approach",** Elsevier
3. Andrew S Tanenbaum, **"Computer Networks"**, Pearson Education
4. Behrouz A Forouzan, **"Data Communication and Networking",** McGraw Hill

Program Core Elective

6th Semester

Course Code: XXXX					Machine Learning					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Understand the basics of Machine Learning
- Explain various Machine Learning Algorithms
- Analyze and design various architectures of Machine Learning algorithms.

UNIT 1 Fundamentals of Machine Learning	**9Hrs.**
Concept of data and their dimensions; Concept of feature and dimension of data; concept of Array, Matrix, and Tensor representation of data; concept of data preprocessing, data imbalance, Outliers, data sampling, feature engineering, Various dimensionality reduction techniques; correlation of features; concept of hypothesis and optimization; concept of classification and regression; Various evaluation Matrix: absolute error, Square error, Sum of square error, Mean Square error, log loss, accuracy, precision, recall, ROC, AUC, F1 score, confusion matrix; concept of Bias and variance .	
UNIT 2: Optimization methods	**7 Hrs.**
Concept of optimization; concept of convex optimization; local minima and maxima; Gradient Descent; concept of learning rate; Stochastic Gradient Descent (SGD); Mini-Batch Gradient Descent; concept of momentum; Adaptive Gradient Algorithm (AdaGrad); RMSProp; Adam Optimization;	
UNIT 3: Clustering, Classification, and regression algorithms	**8 Hrs.**
Partitioning Clustering; Density-Based Clustering; Distribution Model-Based Clustering; Hierarchical Clustering; Fuzzy Clustering; DBSCAN; K-Nearest Neighbors (KNN); Naive Bayes; Decision Tree; Random Forest; Support Vector Machine (SVM); Logistic Regression; XGBoost; Neural Network, Linear regression, Polynomial Regression; Stepwise Regression; Random Forest Regression; concept of boosting and bagging; Concept of overfitting, underfitting, Bias-Variance Tradeoff.	
UNIT 4: Various applications of Machine learning	**4 Hrs.**
Applications of ML in: Healthcare, Finance, Transportation, Manufacturing; energy and environment, Agriculture; Case study: Current Trends and Future of Machine Learning	

Total: 28Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Remember the concept of Machine learning.
CO2: Understand the basic algorithms.
CO3: Apply different algorithms to solve problems.
CO4: Analyze the advantages and disadvantages of various ML algorithms.
CO5: Evaluate the performance of ML algorithms.
CO6: Design an ML algorithm for a specific application.

TEXT/REFERENCE BOOKS

1. Introduction to Machine Learning with Python, Andreas C. MŸller, Shroff/O'Reilly publication.
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems by Aurélien Géron, O'Reilly publication.
3. Deep Learning textbook, Ian Goodfellow, http://www.deeplearningbook.org

24ICxxxT					Cloud Architecture and Services					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Understand the basics of Cloud Architecture
- Explain different types of business and deployment models in cloud.
- Analyze and design different architecture for cloud.

UNIT 1 Fundamentals of Cloud computing and its Architectures	**13Hrs.**
History of computing, Distributed computing, introduction to cloud, virtualization and abstraction, elasticity in cloud, concept of hypervisor, Advantages and limitations of virtual machines, containers, difference between VM and container, virtual networking, Service level agreement.	
UNIT 2: Various architectures of Cloud	**14Hrs.**
Introduction to business models and their architectures, SAAS, PAAS, IAAS, XAAS, Various cloud deployment models and their architectures, Public Cloud, Private cloud, Hybrid Cloud, community cloud, open-source platforms to develop cloud.	
UNIT 3: Data Processing in Cloud	**10 Hrs.**
Load balancing, dynamic and static load balancing, Distributed file system, HDFS, Apache Spark, writing and downloading files in cloud using DFS, data analytics in cloud.	
UNIT 4: Various applications of Cloud	**5 Hrs.**
Cloud based analytics, Cloud based smart applications, Cloud applications in Industry 4.0, Role of cloud in AI applications, Case Study: Design of personal cloud using open-source platforms.	

Total: 42Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Remember the architectures of cloud.
CO2: Understand the basic backbone of cloud.
CO3: Apply different concept of elasticity in cloud.
CO4: Analyze the advantage, disadvantages of various cloud architectures.
CO5: Evaluate the performance of various cloud infrastructures.
CO6: Design the architectures of cloud.

TEXT/REFERENCE BOOKS

1. Cloud Computing: Concepts, Technology & Architecture, by Thomas Erl, Ricardo Puttini, Zaigham Mahmood, PHI publication.
2. Architecting the Cloud: Design Decisions for Cloud Computing Service Models, Michael J. Kavis, Wiley publication.

24ICXXXT					Advanced Web Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

- Learn fundamentals of modern web development.
- To learn client side scripting with ECMAScript.
- Understand the use of MongoDB as an open-source NoSQL database to store and retrieve information.
- Create a Real-time responsive web application using MERN STACK technologies.

UNIT 1: Introduction to UI and UX in Web Development	**6 Hrs.**
Visual elements of User Interface Design, Ideation, Articulation and development in UX Design, Look and Feel/Visual Research, Developing and Refining UI, Sitemap, Wireframes and Prototypes, Fundamentals of HTML, CSS and JavaScript, Pseudo classes, Pseudo-elements, Transitions, Animations, Positioning, Box Model, Bootstrap 4, Material-UI, Basics of Responsive website using media queries, DOM Manipulation, Handling Events using JavaScript, Regular Expressions	
UNIT 2: Front-end Web Development with React.js	**7 Hrs.**
Fundamentals of React.js, Components: Class Components and Functional Components, Basics of Form Handling, Controlled Forms and form Validation, Uncontrolled Components and forms, The Model-View-Controller Framework, The Flux Architecture, Introduction to Redux, React Redux Forms, React Redux Form Validation, Client-Server communication, HTTP Get Request, HTTP Post Request, React Render: Rendering, useState, state immutability, parent and child components, State Management	
UNIT 3 : Server-side Development with Node.js, Express and MongoDB	**10 Hrs.**
Introduction to Server-side Development: Node, Node modules, Node HTTP server, Application Programming Interface, Express Framework and Rest API using Express, HTTPS and Secure Communication, Cross-Origin Resource Sharing, Basic Authentication, Session Based Authentication and Token Based Authentication, JSON Web Tokens, Passport module, OAuth and User Authentication, Backend as a Service, Integrating the React Client and Server, MongoDB and Mongoose	
UNIT 4 : Deployment of MERN Stack Web Applications	**5 Hrs.**
Fundamentals of Secure Deployment, Search Engine Optimization, DNS Record Management, Deploying MERN Stack Application on Web, Free Hosting Platforms, GitHub, Firebase, Heroku	

Total: 28Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand various technologies and trends impacting Single Page Web applications.
CO2 - Understand the use of MongoDB as an open-source NoSQL database to store and retrieve information.
CO3 - Apply a deep knowledge of MVC (Model View Controller) architecture, making the development process easier and faster using open-source technologies.
CO4 - Demonstrate the use of JavaScript to fulfil the essentials of front-end development to back-end development.
CO5 - Implement UI-UX using React.js framework and Responsive web designing for Web Applications.
CO6 - Create a Real-time responsive web application using MERN STACK technologies.

TEXT/REFERENCE BOOKS

1. O'Reilly, “Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node”, Media Publications
3. Chris Northwood, “The Full Stack Developer”, Apress Publications.
4. Greg Lim, “Beginning MERN Stack: Build and Deploy a Full Stack MongoDB, Express, React, Node.js App”,
5. Shama Hoque, “Full-Stack React Projects”, Packt Publishing

24ICxxxT					Image Processing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To foster critical thinking skills by encouraging students to analyze and interpret the results of image processing and computer vision techniques in various applications.
- To demonstrate skills in problem-solving to identify, formulate, and solve image processing and computer vision problems using appropriate methodologies and tools.
- To encourage students to explore interdisciplinary connections by examining usage of image processing.

UNIT 1 Concepts of Digital image processing	**10 Hrs.**
Introduction to image basics, imaging systems, image representation, binary images, grayscale images, color images, multispectral and hyperspectral images, digital image file formats, introduction to image analysis tools, Human Visual Perception and Illusion, Image Fidelity	
UNIT 2 Digital Image Analysis	**10 Hrs.**
Overview of image analysis, preprocessing, region of interest, arithmetic and logic operations, thresholding, connectivity concept, basic object features, segmentation, edge, line detection, Image transforms, DFT, FFT, DCT, WHT, principle component, HAAR transform, filtering. Image enhancement, restoration, compression, registration,	
UNIT 3 : Feature extraction and Analysis	**12 Hrs.**
Feature Extraction, Shape Features, Histogram Features, Color Features, Spectral Features, Texture Features, Feature Analysis, Feature Vectors and Feature Spaces, Distance and Similarity Measures, Data Preprocessing, Pattern Classification algorithms and Methods, Neural networks for image analysis, deep learning,	
UNIT 4 : Applications of image processing	**10 Hrs.**
Medical image processing, Biometric recognition systems, Autonomous navigation, Image translation.	

Total:42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Recognize fundamental concepts in digital image processing
CO2: Comprehend the principles behind image analysis techniques
CO3: Apply various computer vision algorithms to solve real-world problems.
CO4: Evaluate the performance of different image processing algorithms in terms of accuracy, efficiency, and robustness
CO5: Design image processing systems for specific applications to achieve desired functionalities
CO6: Implement image processing algorithms in practical scenarios using functional software solutions.s

TEXT/REFERENCE BOOKS

1. Gonzalez, Rafael C. Digital image processing. Pearson education India.
2. Umbaugh, Scott E. Digital image processing and analysis: human and computer vision applications with CVIPtools. CRC press.
3. Sonka, Milan, Vaclav Hlavac, and Roger Boyle. Image processing, analysis and machine vision. Springer.
4. Dey, Sandipan. Python image processing cookbook, Packt Publishing Ltd.

<Course Code>					Statistical Signal Processing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Introduce mathematical ideas for analysis of random signals and statistical algorithms.
2. Show applicability of statistical analysis in wide range of engineering applications.
3. Explore tools for statistical signal analysis and characterization.

UNIT I: FUNDAMENTALS OF RANDOM SIGNAL PROCESSING	**11 Hrs.**
Review of Statistical Measures and Probability distribution functions, Continuous and Discrete time random processes, Stationarity, Ergodicity, Wide Sense stationarity, Covariance, Correlation and Power Spectral density, Filtering of random processes, Spectral factorization, Wold Decomposition.	
UNIT II: ESTIMATION THEORY	**12 Hrs.**
Maximum Likelihood, Fisher Information, Cramer Rao bound, MAP estimation, MMSE estimation, Non parametric spectral estimation, Periodogram, windowing, Parametric spectral estimation, AR and ARMA modelling, Yule Walker equations, Applications to signal modeling, design of digital communication receivers, channel estimation and forecasting systems.	
UNIT III: ADAPTIVE FILTERING	**12 Hrs.**
Wiener filter, LMS algorithm, recursive least squares, Kalman filters and its variants, Constant modulus algorithm, Adaptive beam forming and generalized side lobe canceller, Applications to noise cancellation, echo cancellation, channel equalization, system identification, linear prediction etc.	
UNIT IV: STATISTICAL SOURCE SEPARATION TECHNIQUES	**07 Hrs.**
The cock tail party problem, Independent component analysis, Blind Source Separation, Partial correlations and concentration matrix, Granger and other causality measures. Applications to speech enhancement, speaker separation etc.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Outline characteristics of random signals, statistical models and algorithms.
CO2	:	Understand classification of statistical signal processing algorithms to be applied in different situations.
CO3	:	Apply statistical signal processing algorithms to various applications like signal estimation and detection.
CO4	:	Analyze and compare performance of statistical signal processing algorithms.
CO5	:	Implement existing statistical signal processing algorithms using modern tools for various engineering applications.
CO6	:	Design a block diagram level simple statistical signal processing system

TEXT/REFERENCE BOOKS

1. Statistical Digital Signal Processing and Modelling, Monson H Hayes, Wiley India.
2. Fundamentals of Statistical Signal Processing: Estimation Theory: Vol. 1, Steven M Kay, Pearson Indian Edition.
3. Modern Spectral Estimation, Theory and Application, Steven Kay, Pearson Indian Edition.
4. Linear Estimation, Thomas Kailath, Pearson Education.
5. Statistical and Adaptive Signal Processing: Spectral Estimation, Signal Modeling, Adaptive Filtering, and Array Processing, Dimitris G. Manolakis, Vinay K. Ingle, Stephen M. Kogon, McGraw-Hill..